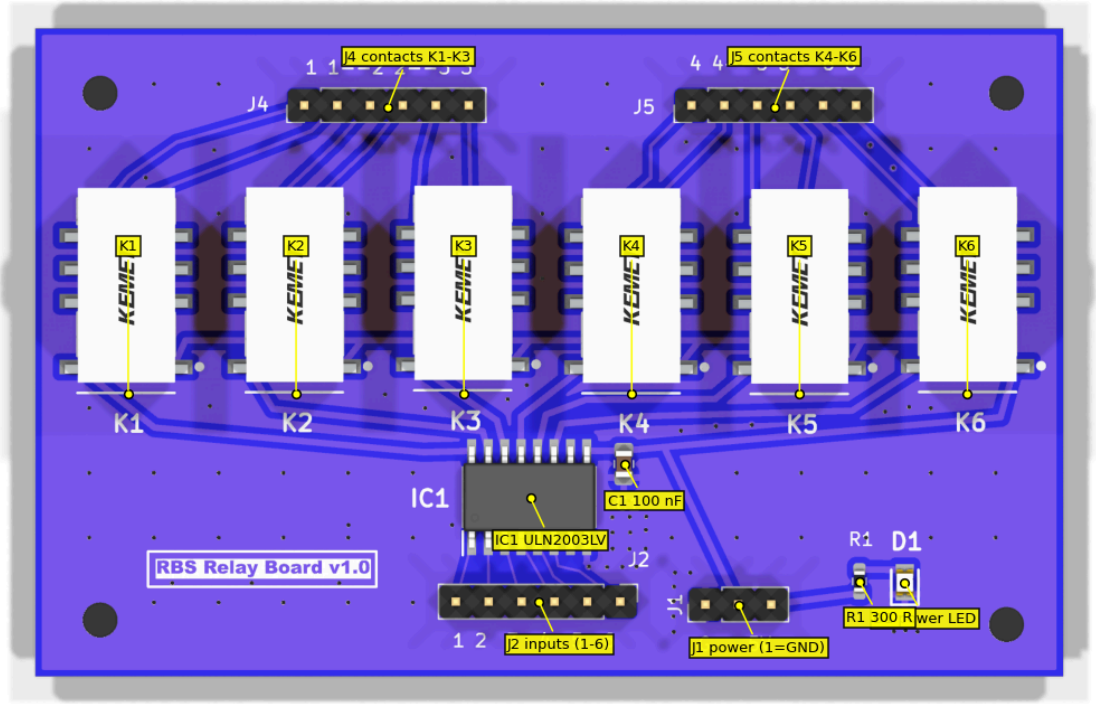
Six-Channel Relay Board

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Six-Channel Relay Board – Engineering and Service Document

Evidence tags: [K] taken from the KiCad project; [D] manufacturer datasheet; [C] calculated here; [I] inferred, not confirmed; [M] needs measurement on hardware. **Nothing in this document was measured on a physical board.**



Annotated top view

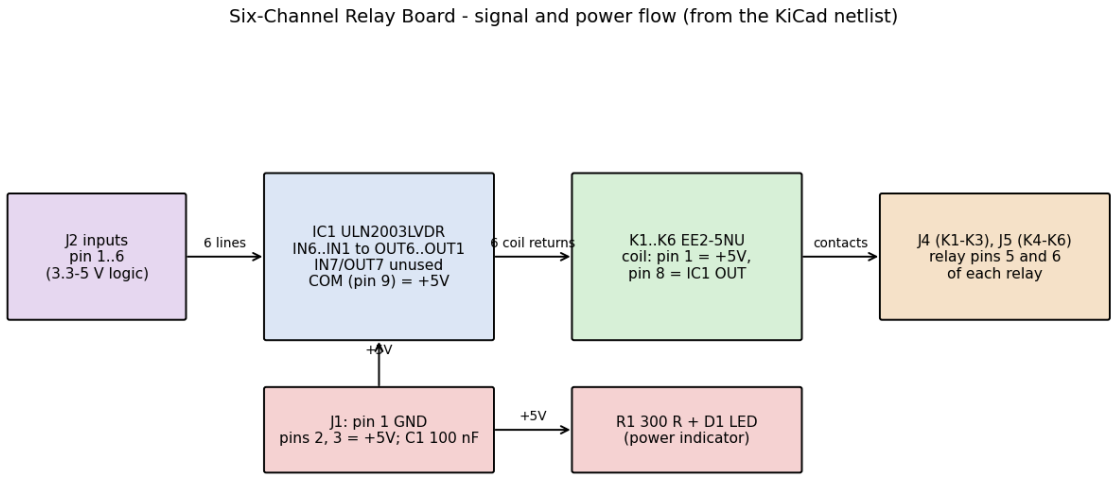
1. Document information

Item	Value
Board	RBS Relay Board v1.0 (silkscreen) [K]
Source	`...\ControlBoardTiny\tinyControlBoard\RelayBoard\RelayBoard` (`RelayBoard.kicad_sch`, `.kicad_pcb`, `production\bom.csv`, `positions.csv`)

Item	Value
Tool	Schematic/PCB files written by KiCad 8.0 format; analysed with kicad-cli 10.0.6
Source dates	PCB and schematic last saved 19 July 2025; production export 30 July 2025 [K]
Revision / date	No revision or date text exists in the schematic title block [K]
Related	Circuit board reference, Open questions, Tiny Control Board

2. Overview

A six-channel, low-side-driven relay board. One ULN2003LVDR driver (IC1) switches the coils of six KEMET EE2-5NU 5 V signal relays (K1–K6). Each relay brings **one contact pair** out to a 2.54 mm pin header (J4 for K1–K3, J5 for K4–K6). Logic inputs arrive on J2; 5 V power and ground arrive on J1. A red LED (D1) with R1 indicates that power is present [K].



Signal and power flow

Not established from these files: what drives J2 and what is connected to J4/J5 in the complete streamer. The link to the Tiny Control Board J6 outputs is a design intent that must be confirmed against that board's documentation.

3. Specifications

Parameter	Value	Tag
Board size	79.375 × 50 mm, two layers, 1.6 mm	[K]
Mounting	4 × M2.5 holes (2.7 mm)	[K]
Supply	Nominal 5 V (coil rating, silkscreen "5V"). The schematic net has no name (auto-named `Net-(IC1-COM)`)	[K][D]
Relay coil	178 Ω ±10 % at 20 °C, operate 3.75 V, release 0.50 V, ≈ 140 mW	[D]
Coil current	≈ 28 mA each at 5.0 V; ≈ 169 mA for all six	[C]
LED current	(5 V − Vf) / 300 Ω; a few mA, Vf not verified	[C]
Driver limits	100 mA/channel at 3.3 V drive, 140 mA at 5 V drive; input absolute maximum 5.5 V; output/COM 8 V	[D]
Driver input thresholds	On ≥ 1.65 V; off ≤ 0.6 V; inputs have internal pull-downs	[D]
Relay contacts	2 A, 30 W / 62.5 VA, up to 220 VDC / 250 VAC switching voltage, minimum load 10	[D]

Parameter	Value	Tag
	mV / 10 µA	
Relay timing	Operate ≈ 2 ms, release ≈ 1 ms	[D]
Coil-to-contact isolation	1500 VAC (relay itself)	[D]
Ambient	Relay −40 to +85 °C; IC −40 to +85 °C	[D]

Margin check [C]: 28 mA per channel is about a quarter of the 100 mA per-channel rating; total 169 mA is far below the 1 A ground-pin limit; IC dissipation is well under the 0.58 W allowed at 85 °C.

4. Connectors, controls and indicators

All connectors are **2.54 mm vertical pin headers**, not screw terminals [K] (this corrects the earlier summary).

Ref	Type	Pin	Function
J1	1×3 header (“Out”)	1	GND
		2, 3	Supply (+5 V), both pins on the same net
J2	1×6 header	1	`/Relay 1 in` → IC1 IN6 → K6
		2	→ IC1 IN5 → K5
		3	→ IC1 IN4 → K4
		4	→ IC1 IN3 → K3
		5	→ IC1 IN2 → K2
		6	`/Relay 6 in` → IC1 IN1 → K1
J4	1×6 header	1–2 / 3–4 / 5–6	K1 / K2 / K3 contact pair
J5	1×6 header	1–2 / 3–4 / 5–6	K4 / K5 / K6 contact pair

Input numbering is reversed relative to the relays: J2 pin 1 operates K6 and J2 pin 6 operates K1 [K]. The net names `/Relay 1 in` and `/Relay 6 in` do not match the relay they drive. Check the harness numbering before use.

Silkscreen: “1 2 3 4 5 6” at J2, “G” and “5V” at J1, “1 1–2 2–3 3” at J4 and “4 4–5 5–6 6” at J5 [K]. Each “n–m” pair marks the two pins of one relay’s contact.

Indicators: D1 (red) lights when the supply is present; it does not show relay state [K]. There are no controls and no dedicated test points [K].

5. Functional description

1. **Power.** J1 pin 1 is ground; J1 pins 2 and 3 supply the board. The supply net feeds IC1 COM (pin 9), pin 1 of every relay coil, R1 and C1 (100 nF to GND) [K].
2. **Drive.** A logic high at a J2 pin turns on the matching IC1 channel. The output pulls relay coil pin 8 low and the relay operates [K][D]. The internal clamp diodes return to COM (the supply), which gives flyback protection [D].
3. **Unused channel.** IC1 IN7 (pin 7) and OUT7 (pin 10) are unconnected. Unused inputs have internal pull-downs [D].
4. **Contacts.** The relay is a 2 Form C part [D]. Only footprint pins 5 and 6 are routed out. By pad geometry these map to datasheet pins 8 and 9 (pole B, normally-open and common) [I]. The expected behaviour is therefore **normally open**: contact closed when the relay is energised. Confirm by continuity test [M].
5. **Mapping.** J2 pin n operates relay K(7−n). The relay’s contact pair appears on J4 (K1–K3) or J5 (K4–K6) as shown in section 4.

6. Assembly and commissioning

Before power [M]: 1. Inspect for solder bridges at IC1 (SOIC-16, 1.27 mm pitch) and the relay pins. 2. Check resistance between J1 pin 1 and pins 2/3 for a short. 3. Check that IC1 pin 1 orientation matches the PCB marking, and the LED polarity.

First power-up (suggested procedure, not a result): 1. Apply 5 V through a current-limited supply; with all inputs low expect only LED current (a few mA). 2. Drive J2 pin 6 high; K1 should click, supply current should rise by about 28 mA [C], and J4 pins 1–2 should close [I]. 3. Repeat for each input in turn and record which contact pair closes. Confirm the reversed numbering in section 4. 4. Operate all six together; expect about 170 mA [C].

7. Testing and troubleshooting

Symptom	Likely cause	Check
LED off	No supply, wrong J1 pin order, R1/D1 fault	Voltage J1 pin 2 to pin 1; D1 orientation
No relay operates	IC1 not powered or ground open; input voltage < 1.65 V	IC1 pins 8 and 9; input level at IC1 [D]
One relay never operates	Open track or solder on IC1 output; failed coil (≈ 178 Ω expected)	Coil resistance between relay pins 1 and 8
Wrong relay operates	Input numbering reversed (by design)	Section 4 table
Relay operates but contact stays open	Contact pair uses datasheet pins 8/9, which are normally open when de-energised	Continuity with relay energised [I]
Relay chatters	Supply sag; no bulk capacitor on the board	Scope the supply at J1 [M]
Contact stays closed	Welded contact after overload, or wrong pole	Measure with the relay off
Input always high	Driving source fault; inputs have internal pull-downs [D]	Disconnect J2 and re-test

8. Repair and rework

- Use a rework station suitable for SOIC-16 for IC1. Check the manufacturer's soldering guidance for the relays before reworking them (not reviewed here).
- Replacement relay: KEMET/YAGEO EE2-5NU (5 V non-latching, 2 Form C). Do not substitute a latching or different-voltage coil.
- Replacement driver: ULN2003LVDR. Other ULN2003 variants have different input thresholds; verify against the datasheet before substituting.
- After any rework repeat the section 6 checks.

9. BOM reference [K]

Ref	Qty	Value	Footprint
C1	1	100 nF	0805
D1	1	LS R976 (red LED)	0805 (LEDC2012X40N)
R1	1	300 Ω	0603
IC1	1	ULN2003LVDR	SOIC-16
K1–K6	6	EE2-5NU	EE2-5NU
J1	1	1×3 header (value field "Out")	2.54 mm vertical
J2, J4, J5	3	1×6 header (value field "Buffer")	2.54 mm vertical
H1–H4	4	M2.5 mounting hole	2.7 mm

The BOM has no supplier part numbers (LCSC column empty). The "Out" and "Buffer" values are labels, not part descriptions.

10. Limitations, risks and open points

Design-check findings (not test results):

1. **Mains-voltage suitability not established.** The relay contacts are rated to 250 VAC [D], but the board uses 2.54 mm pin headers and signal-level spacing. Do not use for mains without a creepage and clearance review [M].
2. **Supply net is unlabelled** and no power flag exists, which gives the ERC warning `power_pin_not_driven` [K].
3. **No input protection or reverse-polarity protection** on J1 or J2. Input drive must not exceed 5.5 V [D].
4. **Decoupling:** one 100 nF capacitor for up to ≈ 170 mA of switching load, no bulk capacitor [C]. Adequacy depends on the supply wiring [M].
5. **Input numbering is reversed** against the relay numbering (section 4) [K].
6. **Contact configuration is inferred** (pole B, normally open) from pad geometry [I]; the second pole is unused.
7. **PCB:** DRC reports 23 warnings and no errors. 20 are silkscreen text-thickness warnings (rule set oddity), 2 are dangling vias on GND and 1 a 0.177 mm dangling track on `Net-(J4-Pin_3)` near (99.9, 51.95) [K]. ERC: 0 errors, 1 power-pin warning, 1 symbol mismatch (LED_Small) and 6 footprint-link warnings that come from the analysis environment.
8. **J1 pins 2 and 3 are the same net** — do not feed two different supplies.
9. The earlier summary page said “one switched contact pair”, “screw terminals” and “details read from a screenshot”. These are superseded by this document.

11. References and revision history

- Texas Instruments, ULN2003LV datasheet, SLRS059B.
- KEMET/YAGEO, EC2/EE2 relay datasheet R7002_EC2_EE2.
- KiCad project files listed in section 1.

Version	Change
1.0	First engineering and service document, written from the KiCad files and datasheets. No hardware measurements.